

Class 11 Biology: High-Yield Diagram Checklist

Chapter 2: Biological Classification (NCERT Page-wise Priority & Labelling Guide)

Instructions for Students: Practice drawing each diagram neatly with pencil and label key morphological/structural parts in CAPITAL letters. Focus particularly on *Very High Priority* figures for Board and competitive exams.

Fig No.	Diagram / Organism	NCERT	Priority	Key Labelling Points / Concepts
Fig 2.6 (a)	Tobacco Mosaic Virus (TMV)	Pg 20	★★★ Top Priority	Helical Capsid (Capsomeres), Single-stranded RNA (ssRNA) core
Fig 2.6 (b)	Bacteriophage	Pg 20	★★★ Top Priority	Hexagonal Head (dsDNA), Collar, Helical Sheath, Tail fibres, Base plate
Fig 2.2	Filamentous Cyanobacteria (<i>Nostoc</i>)	Pg 13	★★★ Top Priority	Heterocyst (N ₂ fixation site), Mucilaginous / Gelatinous sheath
Fig 2.4 (a–d)	Kingdom Protista Types (Dinoflagellate, <i>Euglena</i> , Slime mould, <i>Paramoecium</i>)	Pg 15	★★★ Top Priority	<i>Euglena</i> : Pellicle, 2 flagella. <i>Dinoflagellate</i> : Cellulose plates, 2 flagella. <i>Paramoecium</i> : Cilia, Gullet, Contractile vacuoles
Fig 2.1	Bacteria of Different Shapes	Pg 12	★★☆ High Priority	Cocci (spherical), Bacilli (rod-shaped with spores), Spirilla (spiral with flagella), Vibrio (comma-shaped)
Fig 2.3	A Dividing Bacterium (Binary Fission)	Pg 14	★★☆ High Priority	Cell wall, Cell membrane, DNA / Nucleoid dividing into daughter cells
Fig 2.5 (a–c)	Fungi Morphological Types (<i>Mucor</i> , <i>Aspergillus</i> , <i>Agaricus</i>)	Pg 17	★☆☆ Moderate	<i>Mucor</i> : Mycelium, Sporangiophore, Sporangium. <i>Aspergillus</i> : Conidiophore, Conidia. <i>Agaricus</i> : Cap (pileus), Stipe, Annulus

KEY BOARD EXAM DRAWING & CONCEPTUAL TIPS

- **Viral Structures (Fig 2.6):** Very frequently asked in 2-mark and 3-mark questions. Label the Head, Collar, Sheath, and Tail fibres accurately for Bacteriophage, and helical RNA with Capsomeres for TMV.
- **Nostoc Heterocyst (Fig 2.2):** Remember that the heterocyst is a specialised, thick-walled, anaerobic cell dedicated to atmospheric nitrogen fixation.
- **Five Kingdom Table (Table 2.1, Pg 11):** Thoroughly revise cell wall compositions (Non-cellulosic in Monera, Chitinous in Fungi, Cellulosic in Plantae) alongside the diagrams.